Antoine Hinary

antoine.ihinary@gmail.com +41782350595 Antoine Hinary

Chemin de la Cocarde, 1024 Lausanne, Switzerland

Professional Summary

Robotics engineer specialising in machine learning and computer vision, with an emphasis on taking models from data to deployment. I work comfortably across the full stack, from embedded C/C++ firmware to cross-platform application interfaces, and have led multidisciplinary teams of 60+ engineers to delivery on hard deadlines.

Professional expertise

Programming & Systems

- C++ (real-time implementation)
- Python (algorithm prototyping, data processing, ML pipelines)
- Git, RTDE
- ROS2, serial protocols, hardware interfaces

Perception

- OpenCV & camera calibration
- PyTorch, ONNX, TensorRT (deployment side emphasized)

Machine Learning & Computer Vision

- ONNX model export and graph optimization
- TensorRT inference acceleration for edge deployment
- Latency / throughput trade-off analysis
- Hardware-constrained deployment validation

Embedded & Autonomy Systems

- PX4 firmware development
- MAVLink communication & system integration

Education

Ecole Polytechnique Fédérale de Lausanne (EPFL), *Master in Robotics* 09/2023 – 05/2026
Lausanne, Switzerland

- Specialized in Machine learning, computer vision and automation control.

Ecole Polytechnique Fédérale de Lausanne (EPFL), 09/2020 – 07/2023
BSc in Microengineering Lausanne, Switzerland

- Graduated in the top 15% with a GPA of 5.28, demonstrating consistent academic excellence.

Work Experience

Neurovia Bioelectronics, *Vice President of Product Development* 01/2026 – Present
Geneva, Switzerland

- Designed a multiplexer PCB routing design for a medical In-human cortex stimulation ECoG array.
- Developed STM32 firmware in C for electrode channel switching, real-time fault detection, and hardware safety monitoring.
- Built a cross-platform desktop/web GUI enabling electrode selection across channels and direct serial communication with a Blackrock stimulator.

Grivix, *Master Thesis* 10/2025 – 04/2026
Zurich

- Designed and implemented a complete Machine Learning pipeline for object detection and 6-DoF pose estimation enabling real-time tracking and robotic alignment.
- Generated large-scale synthetic datasets to improve training robustness.
- Built structured training and evaluation workflows using PyTorch.
- Optimized inference latency and throughput using ONNX export and TensorRT acceleration.
- Integrated perception outputs into real-time robotic execution pipelines under latency constraints.

Grivix, *Robotics Engineer* 02/2025 – 10/2025
Zurich, Switzerland

- Led mechanical architecture design in Fusion and actuator dimensioning of a 6-DoF robotic manipulator for industrial operational deployment.
- Defined actuator selection, torque sizing, and motion constraints based on load analysis and application requirements.

Elythor, *Control Automation intern* 06/2024 – 08/2024
Lausanne, Switzerland

- Analyzed and conceptualized a unified controller architecture for a morphing bicopter tailsitter.
- Designed customized trajectory generation algorithms using minimum snap optimization.
- Developed and validated control and simulation models in ROS2-based real-time environments and supported firmware-level implementation.

Projects

EPFL Xplore, *Project Manager & Vice President* 09/2024 – 09/2025

- Led a multidisciplinary team of 60+ engineers to develop a robotic rover and drone platform.
- Coordinated subsystem integration across mechanical, electrical, and software teams.
- Managed timelines and technical milestones.
- Oversaw subsystem validation, testing coordination, and system-level readiness prior to competition deployment.

Morphing Avian Inspired Drone, *Body-Rate Controller* 09/2024 – 01/2025

- Developed custom control laws and actuator allocation logic directly within PX4 firmware (C++)
- Implemented body-rate controller for an 8-DoF morphing aerial platform
- Integrated state estimation and control loops under real-time constraints
- Worked with MAVLink-based communication for system testing and control
- Validated control performance through simulation and hardware testing

Languages

English / Arabic
Bilingual

French
Fluent